



Hazard analysis in the era of AI: Assessing the usefulness of ChatGPT4 in STPA hazard analysis

Stavroula Charalampidou^{*}, Apostolos Zeleskidis, Ioannis M. Dokas

Department of Civil Engineering, Democritus University of Thrace, Xanthi, Greece

ARTICLE INFO

Keywords:
STAMP-STPA
LLM
Hazard Analysis
ChatGPT4

ABSTRACT

This paper investigates the usefulness of LLM, such as GPT4 from OpenAI, in applying the STPA hazard analysis in socio-technical systems. This was achieved by recreating the process of STPA on a verified analysis applied to the ROLFER search and rescue drone system. Firstly, the system was discussed in detail with the LLM, defining its purpose, goals, components, and operations. Then, the LLM was talked through the steps of STPA and prompted to generate UCAs, loss scenarios, and safety specifications. These outputs were compared with those from the human safety team that applied STPA to the same system. The team discovered that LLM can be beneficial in certain aspects, such as loss scenario and safety specification generation and helping the researchers to reach a better understanding of the system, but have drawbacks in others, such as the generation of UCAs and being used as a verification tool of the STPA results of the human analysts.

1. Introduction

Hazard analyses are tools used to ensure the safe operation of sociotechnical systems. Conducting a hazard analysis is a group effort, requiring inputs from domain experts, designers, safety engineers, operators, and others. To collect these inputs while conducting a hazard analysis, groups of meetings are scheduled to understand the system better, exchange ideas, and brainstorm in order to identify potential hazards, their causes, effects and solutions to mitigate their occurrence. This process usually takes many days, even months, considering the rising complexity of the sociotechnical systems and the time availability of experts that need to participate in the meetings.

Today, a new set of technologies has been introduced to the public, namely the Large Language Models (LLMs). LLMs are machine learning models that can perform many different tasks due to them being trained in large data sets (usually taken from the internet) (Wornow et al., 2023). LLMs are being widely used by researchers and professionals in numerous fields to utilize their capabilities in text generation, question –and– answer methods and code generation, to name a few, which makes them eligible to function as models of external experts for complex tasks (Naveed et al. 2023). Regardless of their effectiveness in those domains, LLMs come with certain limitations. Weidinger et al (2022), have categorized the risks associated with the usage of LLM in 6 major categories: I. Discrimination, Hate speech and Exclusion, II. Information

Hazards, III. Misinformation Harms, IV. Malicious Uses, V. Human-Computer Interaction Harms, and VI. Environmental and Socioeconomic harms, with Tamkin et al. (2021) also noticing the biases that ChatGPT-3 exhibits in racial, gender, and religious topics. The same types of biases have also been noted in ChatGPT-4 by Hadi et al. (2023). Until those issues are addressed by the continuous advancements in LLM algorithms and technologies, users should be attentive of them and come to terms with their potential effects.

The literature on the use of LLM in hazard analysis is not extensive, presumably on account of the widespread use of LLM which has emerged during the last year. However, there have been attempts to use Natural Language Processing (NLP) systems, which aim to give computers the ability to process human language to enhance HAZOP by Feng et al. (2021) and Ekramipooya et al. (2023), while Tanguy et al. (2016), Pimm et al. (2012) and NASA (2022), have used NLP to assist in the analysis and classification of safety reports.

Two research teams have started using LLM to enhance the hazard analysis processes within the STAMP accident model. The first approach by Diemert and Weber (2023) was focused on the use of conversational AI tools (mainly OpenAI's Chat-GPT3) to conduct a Co-Hazard Analysis to identify potential factors that may lead to hazardous situations in the context of an STPA analysis. This approach was focused on a simple water heater system and did not comment on the benefits analysts can gain by using LLMs in this way (cost, time savings, etc.). The second

^{*} Corresponding author.

E-mail address: stcharal@civil.duth.gr (S. Charalampidou).

<https://doi.org/10.1016/j.ssci.2024.106608>

Received 26 October 2023; Received in revised form 1 June 2024; Accepted 5 July 2024

Available online 16 July 2024

0925-7535/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

approach by Qi et al. (2023) investigated three ways to utilize ChatGPT-3.5 for STPA hazard analysis. These three ways study differs regarding how hands-on the interactions of the LLM and the human experts are. This research has identified that there is the following gap in the literature “We also believe, there is a pressing need of standardisation on such new way of conducting safety analysis.” The work by Qi et al. (2023) was focused on the first 3 steps of the STPA, especially on the generation and clarification of UCAs: “We also omit the last 2 steps in our later experiments and mainly focus on finding UCAs” and excluded the generation of safety requirements and similarly to Diemert and Weber (2023) didn’t study the potential benefits of LLMs to the hazard analysis process (as aforementioned).

This manuscript aims to investigate in detail the usefulness of a new LLM technology, ChatGPT-4, in the context of an STPA hazard analysis and highlights the strengths and weaknesses of their use, specifically in regard to reducing the time it takes to potentially conduct these analyses by bypassing the need for some of the meetings required between domain experts, enhancing the brainstorming process of identifying hazards, validate the results of previous hazard analysis and identifying safety specifications. This manuscript is differentiated by the literature on using LLMs for STPA on the subject of 1) creating a step-by-step model to conduct the analysis with the help of ChatGPT-4, 2) comparing the time needed to complete the STPA analysis with and without the use of ChatGPT-4 and 3) focusing on Steps 3 and 4 of STPA (meaning UCAs and loss scenarios generation) as well as the production of safety requirements during the entirety of STPA. To achieve this aim in this paper, a comparison is made between a) an application of Steps 3 and 4 of the STPA analysis to a quadcopter Unmanned Aerial Vehicle (UAV) search and rescue system that was conducted without the use of LLM tools (Charalampidou et al., 2020) and b) the same STPA steps are applied to the same system but with the use of a LLM and more specifically the platform of ChatGPT-4.

2. TOOLS

2.1. STPA

STPA, or Systems Theoretic Process Analysis, is an approach to hazard analysis derived from the STAMP (System Theoretic Accident Model and Processes) accident model (Leveson and Thomas, 2018). Using STPA, analysts can determine the specific operational circumstances and contexts in which controllers within a system may initiate unsafe control actions. In this paper, and in the context of STAMP-STPA, safety is defined as: “Freedom from accidents (loss events).” (Leveson, 2016), which excludes reliability, since that is defined “as the probability that something satisfies its specified behavioral requirements over time and under given conditions — that is, it does not fail” (Leveson, 2016).

STPA consists of the following steps:

Step one involves identifying the purpose of the analysis, along with the system boundaries, the system losses, and the system level hazards. In STPA a loss “involves something of a value to the stakeholders. Losses may include a loss of human life or human injury, property damage, environmental pollution, loss of mission, loss of reputation, loss or leak of sensitive information, or any other loss that is unacceptable to the stakeholders” (Leveson and Thomas, 2018), and a hazard is defined as “a system state or set of conditions, that together with a particular set of worst-case environmental conditions, will lead to a loss” (Leveson and Thomas, 2018). Additionally, this step involves defining system-level safety constraints based on these hazards.

Step two focuses on creating a comprehensive model of the system called the Control Structure. The control structure is composed of control loops, as shown in Fig. 1.

The purpose of a control structure is to enforce constraints on the behavior of the overall system. It involves designing a system model that incorporates feedback control loops, along with all the necessary

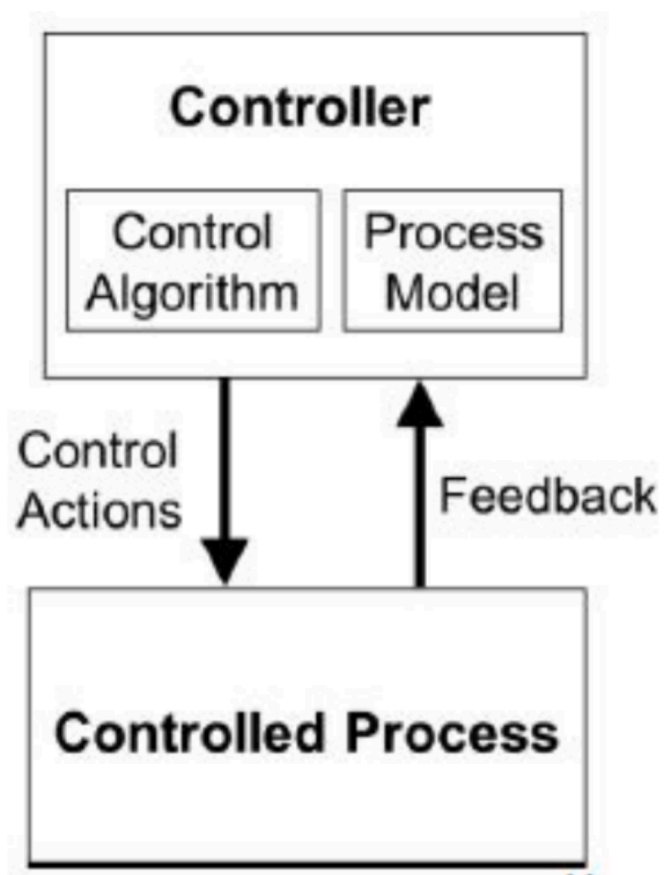


Fig. 1. Generic control loop (Leveson, N.G. and Thomas, J.P., 2018).

components of the system in question, such as controllers, control algorithms, their process models, actuators that enforce the control actions, sensors that provide feedback, and controlled processes.

Step three entails identifying Unsafe Control Actions (UCAs). In STPA, a UCA is defined as: “A control action that, in a particular context and worst-case environment, will lead to a hazard” (Leveson and Thomas, 2018). During this phase, analysts pinpoint system states or operational contexts that may lead to hazards when each control action is a) enforced, b) not enforced, c) enforced too early or too late, d) enforced out of order, stopped too soon, or applied for an extended period of time. The unsafe control actions are typically documented in a table form.

Lastly, step four involves identifying loss scenarios (also known as causal scenarios), meaning the causal factors that can lead to the Unsafe Control Actions and to hazards. There are two categories of identification of the loss scenarios that the analyst must consider, a) scenarios that lead to UCAs and b) scenarios in which control actions are improperly executed or not executed. The loss scenarios can then be used to mitigate the occurrence of the UCAs, implement system changes, and design safer systems.

2.2. ChatGPT-4

ChatGPT-4 is one of the most widely used LLM by the public and industry as of September 2023 (OpenAI, 2023). It is based on the network architecture Transformer (Vaswani et al., 2017), and it is described by its developers as “a Transformer-style model pre-trained to predict the next token in a document, using both publicly available data (such as internet data) and data licensed from third-party providers” (OpenAI, 2023).

The training of GPT models typically follows two stages. First, they

undergo training using a dataset of internet text to predict subsequent words. Following this initial training, the models are fine-tuned using Reinforcement Learning from Human Feedback (RLHF) (Paul et al., 2017) with additional data to generate outputs that align with human preferences. This two-stage process enables language models to acquire capabilities such as few-shot learning and proficiency in a big spectrum of natural language tasks across domains, such as question-answering, arithmetic, and classification (Naveed et al. 2023).

However, it is important to recognize that ChatGPT-4's knowledge is primarily rooted in data up to September 2021 (OpenAI, 2023), with no capacity for learning from post-cut-off events. It may occasionally make simple reasoning errors that do not align with its competency in various domains, and it can sometimes accept false statements from users. It is susceptible to facing challenging problems in ways similar to humans, such as introducing security vulnerabilities into the code it generates. It can also experience "hallucinations" i.e., generate text that is factually incorrect or nonsensical. Additionally, it operates within a limited context window and does not possess the ability to learn from experience. Also, ChatGPT-4 is not without biases in its outputs, meaning the user must take that fact into account while using the generated text.

3. Case study

3.1. The ROLFER system

ROLFER, short for Robotic Lifeguard For Emergency Rescue (Lygouras, Gasteratos and Tarchanidis, 2017), is a fully autonomous UAV system designed and implemented with the primary goal of providing immediate assistance to individuals caught in crisis situations, particularly those at risk of drowning on the beach.

The comprehensive ROLFER architecture consists of the following technical components and operational procedures:

a) The Ground Station: This station comprises a tablet equipped with a dedicated app created by the ROLFER team and connects to a laptop that runs the software which is responsible for the ultimate control of the UAV. Additionally, the ground station includes an RF-telemetry link transmitter–receiver pair and the remote controller for the UAV.

b) The SmartWatch (SW): The SW, accompanied by specialized software for alert signal transmission, operates with a SIM card and communicates via the GSM network. It is waterproof and features GPS functionality.

c) The Fully Autonomous Rescue UAV: The UAV and its associated airborne equipment. The quadcopter UAV component of the system is off the shelf.

An overview of the ROLFER system is presented in Fig. 2.

Further information regarding the technical components of the rescue platform and the Android applications running on it are provided in (Lygouras, Gasteratos and Tarchanidis, 2017, and Lygouras et al., 2018).

The primary human roles within the ROLFER system are divided into the "supervised person" (SP), who is the individual in distress and wears the SW, and the "admin," who serves as the system administrator and is responsible for ensuring the proper functioning of ROLFER and for overseeing the system operations.

During the system setup, the admin defines critical parameters, including reference geographical coordinates for the supervised area, geographical coordinates of potential intermediate points for the UAV trajectory, the phone numbers of the transmission destinations, passwords for both SW and tablet Android applications, the number of SMS messages and the time duration between their transmission to the ground station, and the PIN for SP identification. The admin is also responsible for distributing SW to users and providing instructions on their proper use.

In the event of distress, an SP can activate the HELP button on the SW screen. This action triggers the immediate transmission of a text message containing the person's geographical coordinates via GSM. The message is sent to rescue workers, lifeguards, and the tablet at the ground station. The laptop application calculates the trajectory for the UAV and generates a suitable flight plan file. The UAV's trajectory is determined based on the SP's geographical coordinates, the aerial vehicle's home position, and potential intermediate waypoints, considering obstacles or crowded areas.

Subsequently, the software on the laptop extracts the UAV mission file, and the commands are transmitted to the UAV's open-source

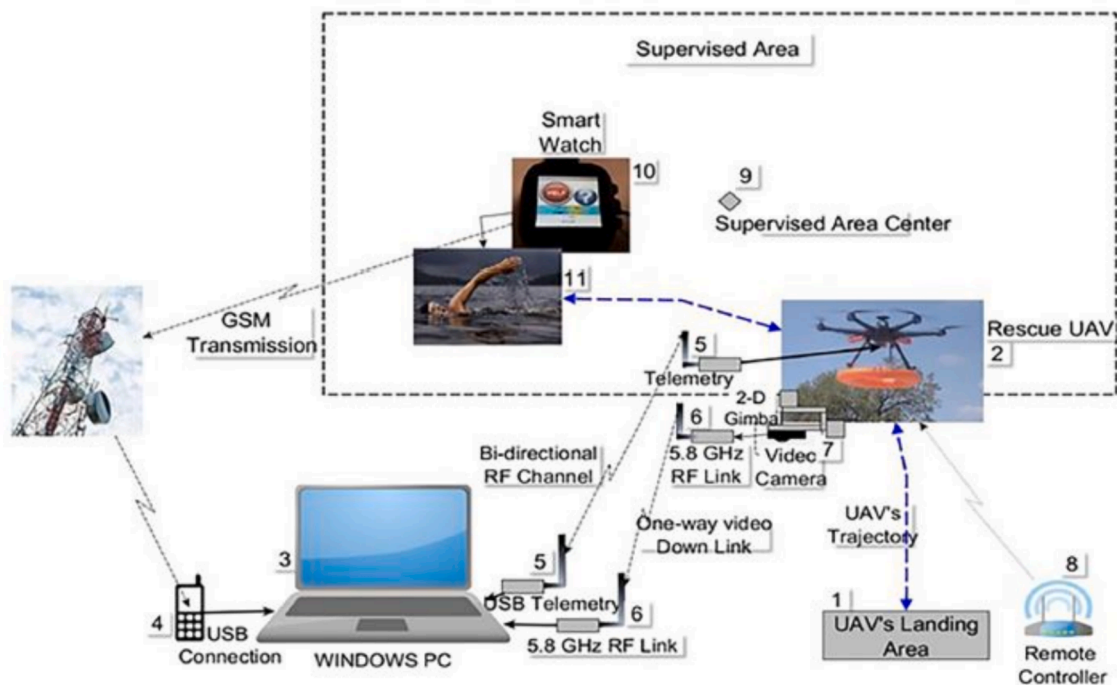


Fig. 2. The ROLFER system (Lygouras et al., 2018).

autopilot. This process promptly activates the fully autonomous aerial vehicle's rescue mission. ROLFER's UAV can operate and navigate without its remote controller (RC) since it relies solely on data from the SP's wearable equipment. Upon reaching the SP's location, the UAV descends to a predetermined altitude, typically three meters, and releases the rescue apparatus to the distressed individual. It then ascends to another predetermined height, usually ten meters, and autonomously returns to a designated landing area, safely situated away from beachgoers and the base station, marked and fenced for the safety of individuals and equipment.

To conduct the analysis, four three-hour meetings in two months were done with the system's developers to first understand the system and its components and second to present the results of the STPA analysis and discuss their validity and the ease of the applications of the suggested safety specifications. The STPA analysis was concluded in a period of 4 to 5 weeks due to new information being disclosed by the developers, necessitating the implementation of those into the analysis. The control structure changed three times before reaching its final form and being approved by the developers. The UCA table was also changed during that time, implementing the new information.

3.2. Method

In brief, the method followed was divided into five steps. The first step aims to acquaint ChatGPT-4 with the system under study, which is achieved by describing the system and then answering potential questions that ChatGPT-4 might need to comprehend better how the system operates. Steps 2 to 5 follow a similar process, with their aim being to provide all the necessary information that the ChatGPT-4 needs to follow so that it can complete Steps 3 and 4 of an STPA analysis. This is achieved by defining the parameters STPA needs as described by the literature and then requesting ChatGPT-4 to generate such parameters for the system in question. For example, in Step 3, the description of what UCAs are and what the correct way to generate them is provided to ChatGPT-4, and then the LLM is requested to produce the UCAs for the system under study. This process is described in detail in the following sections but can also be seen in Fig. 3.

Step 1: Define the system.

Firstly, the system in which the human team conducted the hazard analysis was described briefly to ChatGPT-4. This was done without

referencing that a hazard analysis would occur or that the LLM should act as a Safety Engineer. After the brief description of the system, a complete description of the system and its operation was provided. To achieve that, the system and software description, the technical information about the components of the system and the way they interact with each other, as well as the control structure of the system, were taken from the published research about the ROLFER system and prompted into the LLM chat exchange. Since at the time that this paper was authored, it was not possible to prompt images or diagrams in the chat exchange, the control structure of the system was described by explaining the connections between the system components as followed: "The admin can give the Return to Land command to the UAV and in order to do so, it needs the feedback of flying trajectory plan, GPS coordinates, Battery level". This was repeated until both control structures of the system were explained to the LLM. Since the focus of this paper was on the Steps 3 and 4 of the STPA analysis, the LLM was not used to aid in the generation of losses and hazards, as well as the generation of the control structure.

After completing the system description, ChatGPT-4 was asked if it needed more information to comprehend the system and how it operates better. It provided sets of questions to be answered by the analysts and generated new questions based on the answers the researchers provided it with (a sample of those questions is presented in Table 1). This continued until the answers for the last nine questions of ChatGPT-4 were "No – we have not yet implemented any measures to mitigate the dangers," which led to the LLM not generating more questions about the system's operation.

Step 2: Defining the boundaries of the analysis

Secondly, by using the STPA handbook (Levenson and Thomas 2018), the exact definitions and explanations of what losses and hazards are in the context of the STAMP accident model were prompted, and afterwards the losses and the hazards of the system in study, that the human analysts generated, were provided to ChatGPT-4. Also, the guidelines about the generation of system-level safety constraints were taken directly from the STPA handbook (Levenson and Thomas 2018) and were prompted into the LLM, followed by a request to generate those for the hazards that were provided for the system. The definition of controllers, control actions, unsafe control actions, the methodology for producing UCAs, and the structure of the UCA table were also provided.

Step 3: Unsafe Control Actions UCAs.

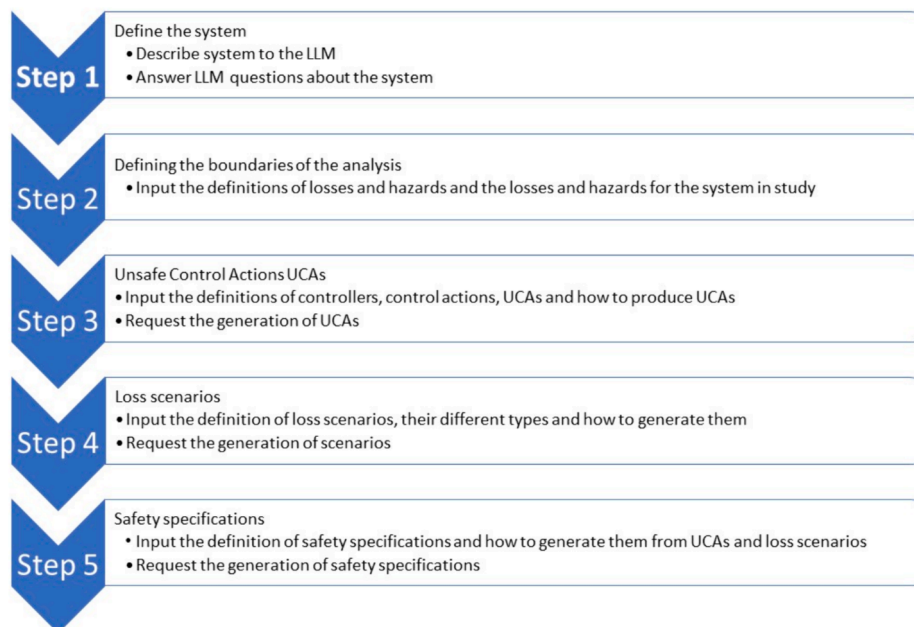


Fig. 3. Diagram of the proposed steps followed to conduct an STPA hazard analysis using ChatGPT-4.

Table 1
Indicative ChatGPT-4 questions and analysts' answers.

Given that the base station and the UAV's base will be secured, could you explain more about the security measures that will be implemented to protect these components from potential human attacks or sabotage?
<i>Since the UAV needs a landing and docking area with specific dimensions, a metal fence with poles will be built around it. The base station will always have someone in attendance while in operation, and its components will be secured while the system is not operating.</i>
While it's understood that the user can cancel a rescue operation, what happens if the user is unconscious or otherwise unable to cancel the operation but has moved out of the supervised area after pressing the help button?
<i>The UAV stops and returns to land, but the SW keeps transmitting the new coordinates at the scheduled intervals and alerts the SaR operations.</i>
Is the system equipped to handle multiple emergencies simultaneously? If more than one user sends a distress signal at the same time, how does the system prioritize who to help first?
<i>It is a first come, first serve system since there is only one drone available at the moment.</i>
Regarding the UAV's navigation and GPS tracking, how does the system handle situations where the GPS signal is weak or momentarily lost?
<i>The system uses the last known coordinates of the person in need and deploys the life ring there, regardless of the current position of the user.</i>

Thirdly, ChatGPT-4 was instructed to generate UCAs for each control action of the system in study. The control actions were prompted individually, and the LLM provided a UCA table for each of them. When it provided the UCA table for the first control action, it was observed that it generated exactly 7 UCAs under each category, so for the second control action, it was prompted to generate less than 7 UCAs under each category, in order to avoid the LLM from answering the rest of the prompts using the same arbitrary pattern, which it did as instructed, since by this point the first signs of the LLM using patterns in its answers had emerged. This sequence of prompts and responses continued until all the desired control actions were prompted to ChatGPT-4 and the corresponding UCAs were generated.

Step 4: Loss scenarios.

Lastly, the definition of what a loss scenario is in the context of an STPA analysis, the different types of loss scenarios that can be generated based on STPA (i.e., unsafe controller behaviour, inadequate feedback, etc.) and the instructions about how to generate them were prompted exactly as referenced in the STPA handbook, omitting only the examples given in the book with the aim of not confusing the LLM by referencing a different system than the one studied at the time. The UCAs created by the human analysts were then prompted, and the LLM was asked to generate the loss scenarios referring to those UCAs. This procedure was completed with no issues.

Step 5: Safety specifications

After the completion of the loss scenarios generation, ChatGPT-4 was prompted to generate safety specifications that would mitigate the occurrence of those loss scenarios.

4. Results

In the following section, the results of the responses to the prompts of the steps described in the above section will be presented together with statistics about them.

During Step 1, after being asked if it needed more information about the system, ChatGPT-4 asked 35 questions to “understand” the system and how it operates at any given time. These questions were categorized as a) questions regarding the mechanical parts of the system and b) questions regarding the humans and their way of interacting with the system and each other. A subset of those questions is shown in Table 2.

The last group of 9 questions provided by the LLM were answered with “No – we have not yet implemented any measures to mitigate the dangers.” Four of these questions mentioned potential issues that the system might face; however, these issues had been defined by the UAV

manufacturer and the safety team as having no immediate effect on the system and were outside the boundaries of the hazard analysis that took place. An example of this group of questions is provided in Table 3.

The second reason for providing the “No – we have not yet implemented any measures to mitigate the dangers” answer to ChatGPT-4 was because 5 of LLM’s questions identified issues correctly, but the team did not want to reveal the mitigating specifications defined when the team did the analysis of the system. The goal was to see if ChatGPT-4 could define the proper safety specifications for these issues after applying the steps of STPA analysis. In essence, the team did not want to reveal the safety specifications they already knew to the LLM.

For example, one question of this category was: How often does the admin need to perform checks on the system? Are there any automated alerts or notifications for potential system issues?

By not informing ChatGPT-4 about that at this stage of the process, it was able to generate the need to implement new feedback channels as a safety specification for the system after implementing the steps of STPA. This process can be seen in Table 4.

After the LLM ceased asking questions about the system, it provided four suggestions, shown in Table 5, as potential improvements to the system without being prompted to do so.

In Step 2, ChatGPT-4’s responses to the definition of the parameters needed for the STPA analysis were the typical text it provides, acknowledging it received the information that the user has prompted it with.

Afterwards, it was prompted to generate the system level safety constraints based on the information received, generating 8 of them (see Table 6).

The first five were paraphrased from the five system hazards, and the last three were again paraphrased from the last three system hazards, with the addition of the phrase “*must be detected and measures taken to*” continued by a suggested mitigation. This phenomenon, the paraphrasing of the prompt as an answer, was noted throughout the answers of ChatGPT-4 from this point onwards.

In Step 3, ChatGPT-4 was prompted with 18 control actions, one at a time, to generate the UCAs for these control actions. The prompt for the first control action included the phrase “generate a maximum of 7 UCAs under each category”, but after noticing that the LLM generated exactly 7 UCAs for each category, the prompt for the control actions 2 to 18 was changed to “generate less than 7 UCAs under each category”. Despite being prompted to generate as many UCAs as possible for those control actions, it kept generating a maximum of 4 UCAs under each category. ChatGPT-4 generated a total of 138 UCAs for 18 control actions. It was

Table 2
Categorisation of ChatGPT-4 questions.

Questions regarding the mechanical parts
• The UAV and the system components may be exposed to harsh summer elements like sand and salt, which can potentially damage the equipment. Is there any plan to safeguard the system and prolong its lifespan in such conditions?
Regarding the UAVs navigation and GPS tracking, how does the system handle situations where the GPS signal is weak or momentarily lost?
Questions regarding the humans and their way of interacting with the system and each other.
What kind of training or instructions might the individual need to make use of the delivered aid? Are there specific procedures for different kinds of emergencies?

Table 3
Questions identified as “No immediate effect”.

<i>Given that the system is designed to operate in summer, how does it handle extreme heat conditions? Can prolonged exposure to sunlight or high temperatures affect the system’s performance or lifespan?</i>
<i>How does the system handle potential signal losses or communication disruptions, either between the UAV and the base station or between the smartwatch and the base station?</i>
<i>The UAV and the system components may be exposed to harsh summer elements like sand and salt, which can potentially damage the equipment. Is there any plan to safeguard the system and prolong its lifespan in such conditions?</i>
<i>When the UAV lowers its height to 3 m to release the life ring, is there a mechanism to ensure that the UAV maintains a safe distance from large waves or other potential obstacles at sea level?</i>

Table 4
Safety specification generation from ChatGPT-4.

UCA 53: Admin stops charging UAV battery prematurely
Loss scenarios:
The admin misreads the battery charge level and mistakenly believes the battery is fully charged.
Lack of feedback or delays in receiving feedback on battery status cause the admin to believe the battery is fully charged and stop the process prematurely.
Safety specifications
Implement real-time, accurate feedback systems for monitoring battery charge status, and train admin to correctly interpret this feedback.

Table 5
Suggestions for improving the system.

Developing a more robust fail-safe mechanism or backup plan to ensure aid reaches the distressed individual even if the UAV has to abort the mission.
Considering mechanisms for informing users about the status of their distress signal, especially if the UAV is currently engaged in a rescue mission.
Implementing protective measures to safeguard the system components against harsh environmental elements, which can extend the lifespan of the system.
Introducing protocols for regular system checks, or even better, automated alerts for potential system issues, which could improve the system’s reliability and performance.

Table 6
System level safety constraints generated by ChatGPT-4.

SC1: The ROLFER system must be adequately prepared for its mission at all times to prevent the loss of system goal and public trust [H1].
SC2: A Supervised Person (SP) must be adequately served during the phase of preparation and handling of the Smartwatch (SW) to maintain public trust [H2].
SC3: The UAV must successfully approach the distressed human under all circumstances to prevent loss of life and property [H3].
SC4: The UAV must maintain a minimum safe separation distance from persons or objects/obstacles on the beach or the water level [H4].
SC5: In the event of the UAV approaching the SP, it must successfully provide rescue assistance to prevent loss of life [H5].
SC6: If the UAV cannot approach the distressed human, then the failure must be detected and alternative measures must be taken to prevent loss of life [H3].
SC7: If the UAV is close to a person or object/obstacle, then the proximity must be detected and measures taken to prevent a collision or harm [H4].
SC8: If the UAV approaches the SP but cannot provide rescue assistance, then the failure must be detected and measures taken to ensure that the person is rescued [H5].

noted that the LLM did not link the UCAs to the corresponding hazards until asked specifically to do so, and even then, the majority of the links were incorrect.

Afterwards, ChatGPT-4 was prompted with the assumptions made by the human team in the original analysis and was then asked to generate a new UCA table, considering these assumptions. The new UCA table consisted of 17 UCAs for the 18 control actions, with two control actions having 2 UCAs for each control action, three control actions having 0 UCAs and 13 control actions having only 1 UCA in the not provided category. These UCAs all shared the same format of “controller fails to do {control action}.”.

5. Validity of results

The first set of 138 UCAs were placed into four categories based on their validity: correct (69 UCAs), incorrect, i.e., provided false context, were not applicable to the system or did not follow the guidelines provided (31 UCAs), in the wrong place i.e., UCAs that when moved to another column were correct (20 UCAs) and UCAs that need more context or needed to be developed further in order to be categorized (18 UCAs).

It was observed during this time that after the 11th control action and until the final 18th control action, the UCA table that ChatGPT-4 provided had developed a pattern. More specifically, there were always 2 UCAs in the provided category, 2 UCAs in the not provided category, 3 UCAs in the wrong timing/wrong order category, and 1 UCA in the stopped too soon category. Out of those, one UCA from the wrong timing/wrong order category was always in the wrong place and needed

to be in the provided category. A similar pattern was noted for the control actions with 8 UCAs in total, with the pattern being 2 in the provided category, 2 in the not provided category and 4 in the wrong timing/wrong order category. Also, in the “not provided” category, ChatGPT-4 used the word “failed to” almost in every UCA. It is worth mentioning that 4 of the 138 UCAs generated referred to information about the system that was not prompted by the analysts (hallucinations) and were false.

In comparison, the human analysts generated 57 UCAs for the same 18 control actions. The UCAs generated by human analysts and by ChatGPT-4 were then compared. Of the 57 UCAs from the human analysts, 46 were also generated by the LLM in some form. It was noted that while the human analysts created separate UCAs for different system checks, the LLM would use only one UCA referring to all the checks, as shown in Table 7.

The majority of the UCAs that ChatGPT-4 did not generate and the

Table 7
UCAs generated by humans and ChatGPT-4.

Control action: check the rescue equipment	
ChatGPT-4	Human analysts
UCA: Admin neglects to verify the functionality of the rescue equipment	UCA: Admin did not check the rescue equipment, while the life ring was damaged
	UCA: Admin did not check the rescue equipment while there was a problem with the life ring release mechanism
	UCA: Admin did not check the rescue equip, met while the life ring was not attached to the UAV

human analysts did, were about the correct order of the procedures that should take place in the system, in the “wrong order” column.

In step 4, ChatGPT-4 was asked to generate loss scenarios for 6 UCAs derived from the human analysts. The format it followed was exactly the same as the one in the prompt about the way to generate loss scenarios, e.g., controller-related failures, inadequate control algorithm, unsafe control input, and inadequate process model. When asked to provide loss scenarios about a specific control action, it generated scenarios for those five categories:

- Failures involving the controller
- Inadequate control algorithm
- Unsafe control input
- Inadequate process model
- General specifications

It was noted that in every case, both for UCAs and Control Actions, ChatGPT-4 generated a maximum of 3 loss scenarios for each category, with averaging on two loss scenarios for each category after the first five prompts. The LLM made assumptions about how a specific part of the system worked and provided loss scenarios based on those, for example, providing loss scenarios involving the SW distribution to users while assuming that the admin would use a software to facilitate the distribution.

In Step 5, it was requested from ChatGPT-4 to generate safety specifications for the loss scenarios it had generated in the previous step. The LLM provided only one safety specification for each loss scenario. While the safety specifications did directly correlate with the loss scenarios and made sense in this regard, the validity of the specifications depended on the validity of the loss scenarios. For instance, since the “*The automated system fails to charge the UAV battery before preparation and the admin has to manually charge it after the preparation*” loss scenario is wrong, based on the fact that there is no such automated system in the system in study, the safety specification “*Update the automated system to detect and signal true battery overheating or charging errors more accurately*” is also wrong as the one is the product of the other.

6. Conclusions and discussion

In this paper, the authors investigated the usefulness of ChatGPT-4 when conducting an STPA hazard analysis. This was achieved by recreating the process of conducting an STPA analysis with ChatGPT-4 on a previously verified STPA on the ROLFER system.

Based on the fact that out of the 138 UCAs that ChatGPT-4 generated, 69 of those were either incorrect, lacking context, or placed in the wrong category, as shown in [section 5](#), ChatGPT-4 does not provide UCAs in a satisfactory manner (1 in two are not accepted). Thus, ChatGPT-4 results should always be validated by STPA experts. This is probably the biggest challenge of this approach, because an analyst with not much experience in the application of STPA will not be able to rely on ChatGPT-4 to validate their results or even use the LLM as inspiration for more UCAs and whole process of the generation of the UCA table from the LLM and the subsequence validation of it does not save enough time to justify the usage of it by an experienced analyst. It must be noted though, that ChatGPT-4 generated a correct UCA that the human analysts did not, leading to a new and correct safety specification.

At the end of the application of each step of STPA, the need to translate the results into safety specifications emerges. For example, from the UCA “Admin connects the battery before ensuring it is fully charged,” a safety specification that can be generated is that “the admin must connect the battery to the UAV after ensuring that the battery is fully charged.” The same procedure has been followed for the generation of safety specifications from the scenarios in Step 4. ChatGPT-4 can easily and quickly generate these safety specifications.

It was noted though, that the capabilities of ChatGPT-4 were diminishing with each message exchange. This was made apparent due

to errors in numbering, multiple spelling and grammar errors, and the fact that ChatGPT-4 followed specific patterns in its answers. The grouping of information to condense the chat exchanges should be investigated as potentially being able to be used to mitigate that deterioration.

Despite these issues, ChatGPT-4 can be a rather powerful tool in safety analysis. The the questions it posed about the system can serve as valuable tools for analysts to significantly reduce the time required for brainstorming and consulting with system developers. By offering these questions to the system developers and requesting their answers, an almost perfect understanding of the system and its operational mechanisms can be attained. ChatGPT-4 can also be used for brainstorming and for bouncing ideas off of. While the UCAs generated could not be used to validate the results of the human analysts, many of the loss scenarios that were generated were correct and can be used. The wrong loss scenarios or those that did not apply to the system, were easy to differentiate and not taken into account, compared with the identification of the wrong UCAs, which might have proved difficult for a non-seasoned analyst.

Since the training of the ChatGPT-4 about the system in study consisted of using the published literature about the ROLFER and then answering the questions that the LLM provided, the training was completed in a few hours. The training on the usage of the STPA was also very fast, being completed in under 1 h, using the STPA handbook. The generation of the UCAs for each control action was almost instantaneous, same as the generation of loss scenarios and safety specifications, bringing the total time to complete the STPA analysis to less than 8 h. The results validation, categorization, and comparison with the analysts’ results took about 8 h. In comparison, the complete analysis was completed by the analysts in a period of 4 to 5 weeks, as described in detail in the [section 3.1](#).

By using LLM tools such as ChatGPT-4, the total time needed for a complete execution of an STPA analysis in a complex sociotechnical system can be greatly decreased due to the generation of questions that help with the understanding of the system, and the generation of the causal scenarios and safety specifications. ChatGPT-4 can also be used to check for duplicate or similar safety specifications and UCAs, a rather tedious and time-consuming procedure.

7. Future work

Plans for future research on this topic include but are not limited to, recreating this process using other LLM tools such as Google Bard, Microsoft Bing Chat, Jasper.ai, etc., using the same process on different analyses and safety-critical subsystems to validate the proposed methodology for using LLM in STPA. Also, the usage of a reduced and simplified set of prompts to achieve a shortened text exchange with the LLM, as the findings of this paper showed that the length of the text exchange affects the capabilities of these LLMs, is another aim to investigate as future work.

CRediT authorship contribution statement

Stavroula Charalampidou: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Apostolos Zeleskidis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Ioannis M. Dokas:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Charalampidou, S., Lygouras, E., Dokas, I., Gasteratos, A., Zacharopoulou, A., 2020, September. A Sociotechnical Approach to UAV Safety for Search and Rescue Missions. In: 2020 International Conference on Unmanned Aircraft Systems (ICUAS) (pp. 1416-1424). IEEE.
- Diemert, S., Weber, J.H., 2023. Can Large Language Models assist in Hazard Analysis? arXiv preprint arXiv:2303.15473.
- Ekrampipooa, A., Boroushaki, M., Rashtchian, D., 2023. Application of Natural Language Processing and Machine Learning in Prediction of Deviations in the HAZOP Study Worksheet: A Comparison of Classifiers. *Process Saf. Environ. Protect.*
- Feng, X., Dai, Y., Ji, X., Zhou, L., Dang, Y., 2021. Application of natural language processing in HAZOP reports. *Process Saf. Environ. Protect.* 155, 41–48.
- Hadi, M.U., Qureshi, R., Shah, A., Irfan, M., Zafar, A., Shaikh, M.B., Akhtar, N., Wu, J., Mirjalili, S., 2023. A survey on large language models: Applications, challenges, limitations, and practical usage. *Authorea Preprints*.
- Leveson, N.G., Thomas, J.P., 2018. *STPA handbook*. MA, USA, Cambridge.
- Leveson, N.G., 2016. Engineering a safer world: Systems thinking applied to safety (p. 560). The MIT Press.
- Lygouras, E., Gasteratos, A., Tarchanidis, K., 2017. ROLFER: An innovative proactive platform to reserve swimmer's safety. In: *Information Systems for Crisis Response and Management in Mediterranean Countries: 4th International Conference, ISCRAM-med 2017, Xanthi, Greece, October 18-20, 2017, Proceedings 4* (pp. 57-69). Springer International Publishing.
- Lygouras, E., Gasteratos, A., Tarchanidis, K., Mitropoulos, A., 2018. ROLFER: a fully autonomous aerial rescue support system. *Microprocess. Microsyst.* 61, 32–42.
- NASA, 2022. Natural Language Processing Techniques for Intelligent Knowledge Management of Safety Reports (online). Available from: <https://ntrs.nasa.gov/citations/20220016425> (Accessed 30 January 2024).
- Naveed, H., Khan, A.U., Qiu, S., Saqib, M., Anwar, S., Usman, M., Barnes, N., Mian, A., 2023. A comprehensive overview of large language models. arXiv preprint arXiv: 2307.06435.
- OpenAI 2023 GPT-4 Technical Report Available online: <https://cdn.openai.com/papers/gpt-4.pdf> (last checked 29/09/2023).
- Christiano, Paul F., Leike, Jan, Brown, Tom, Martic, Miljan, Legg, Shane, Amodei, Dario, 2017. Deep reinforcement learning from human preferences. *Advances in Neural Information Processing Systems*, 30, 2017.
- Pimm, C., Raynal, C., Tulechki, N., Hermann, E., Caudy, G., Tanguy, L., 2012. Natural Language Processing (NLP) tools for the analysis of incident and accident reports. In: *International Conference on Human-Computer Interaction in Aerospace (HCI-Aero)*.
- Qi, Y., Zhao, X. and Huang, X., 2023. safety analysis in the era of large language models: a case study of STPA using ChatGPT. *arXiv preprint arXiv:2304.01246*.
- Tamkin, A., Brundage, M., Clark, J., Ganguli, D., 2021. Understanding the capabilities, limitations, and societal impact of large language models. arXiv preprint arXiv: 2102.02503.
- Tanguy, L., Tulechki, N., Urieli, A., Hermann, E., Raynal, C., 2016. Natural language processing for aviation safety reports: from classification to interactive analysis. *Comput. Industry* 78, 80–95.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. *Advances in neural information processing systems*, 30.
- Weidinger, L., Uesato, J., Rauh, M., Griffin, C., Huang, P.S., Mellor, J., Glaese, A., Cheng, M., Balle, B., Kasirzadeh, A. and Biles, C., 2022, June. Taxonomy of risks posed by language models. In: *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency* (pp. 214-229).
- Wornow, M., Xu, Y., Thapa, R., Patel, B., Steinberg, E., Fleming, S., Pfeffer, M.A., Fries, J., Shah, N.H., 2023. The shaky foundations of large language models and foundation models for electronic health records. *NPJ Digital Med.* 6 (1), 135.